

to use another IDE, you can definitely do so. However, you will need to manually handle the configuration, integration, and updates yourself, instead of letting Beacon Mountain do it automatically. Android NDK is the Native Development Kit that supports implementation of Native C/C++ code in your Android applications. Cygwin is another important utility that provides a Unix*/Linux*-like environment to interface with tools such as vi, make, gcc, etc. This becomes important when trying to implement code that uses both Java and C++ in the underlying source code. Since it is listed as a requirement for the Android NDK, Beacon Mountain tools bundle Cygwin as well.

Beacon Mountain also includes all the Android Design elements that are important for designing the UI of most Android applications. This includes icon packs, stencils, Robotofont, etc. Google also provides these elements in the Adobe Photoshop* and Illustrator formats, so that they are easy to integrate with the rest of the applications and can be modified accordingly. Google recommends that you design your application according to the Holo Design Guidelines. The Holo interface is the theme generally adhered to while designing most Android apps since Android Honeycomb 3.0. The three available Holo themes are Holo Dark, Holo Light, and Holo Light with dark action bars. You may refer to the online design guide for more details here: <http://dgit.in/holodesign>.

INTEL SOFTWARE MANAGER

The Intel Software Manager is a utility installed by Beacon Mountain. It is available for all three operating system environments (including Linux), although it is only distributed for Windows and OS X in combination with Beacon

Mountain. Intel Software Manager downloads and installs updates for all software. Intel Software Manager allows you to update subscriptions and serial numbers as well as retrieve the latest news related to Intel software development products. You can configure various parameters related to software updates using its interface. This includes the frequency at which updates are checked, download location, notification method, and other privacy settings. The update frequency can be set at manual, when an Intel tool is launched, at system start-up, or on a periodic time-based frequency. The default download location of updates for Intel Software Manager is as follows:

- Windows OS: [MYDOCUMENTS]\Intel\Download
- Linux OS:

Root (sudo) users: /root/intel/download

Non-root users: \${HOME}/intel/download

- OS X:

Root (sudo) users: /root/intel/download

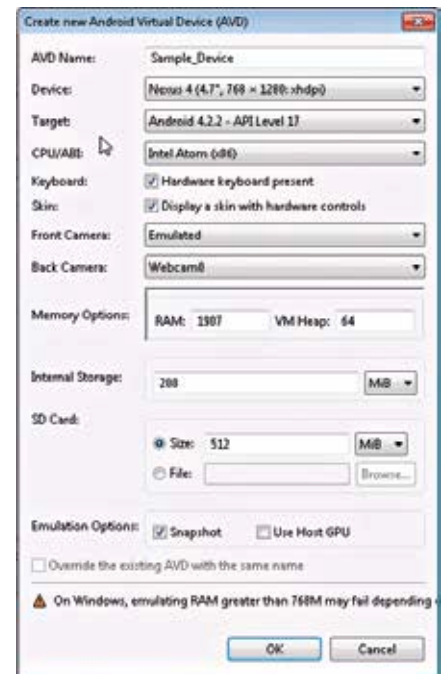
Non-root users: /Users/[user]/intel/download

INTEL® HARDWARE ACCELERATED EXECUTION MANAGER (INTEL® HAXM)

The Intel HAXM is a virtualization engine or hypervisor that utilizes the Intel hardware-assisted virtualization technology to greatly speed up the execution of the Android App emulator. Intel HAXM supports Microsoft Windows, Apple OS X, and Linux (Ubuntu*). However, it is distributed as part of the Beacon Mountain bundle for Microsoft Windows and Apple OS X only. Linux users can set it up manually using KVM virtualization with the help of the installation tutorial available online here: <http://dgit.in/inteltuto>.

The minimum requirement for Intel HAXM is an Intel processor that supports Intel® Virtualization Technology (Intel® VT) or hardware-assisted virtualization, EM64T and Execute Disable (XD) Bit functionality along with at least 1 GB of RAM. Beacon Mountain checks for all these requirements during installation and will notify you accordingly if they are not met. If you are sure that your processor supports this functionality and you still get this error, then you should check your BIOS to see if the Hardware Virtualization

feature has been turned off. Once the installation is complete, you can check if Intel HAXM is running or not by executing the "sc query intelhaxm" command on Microsoft Windows or the "kextstat | grep intel" command on Apple OS X. One important thing to understand is that Intel HAXM only supports Intel Atom processor-based x86 images for the emulator.

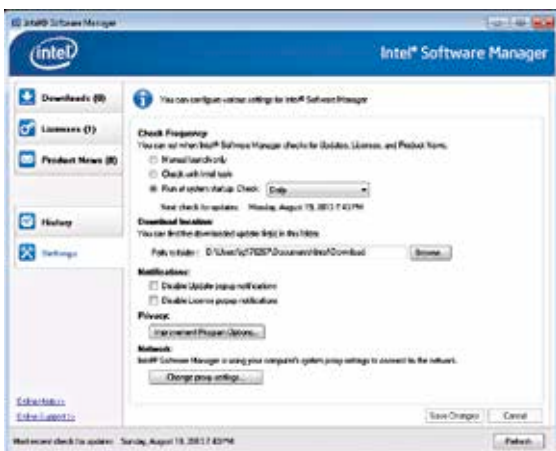


Creating an Android x86 virtual device

To create this image, you can choose the CPU/ABI as "Intel Atom (x86)" and a RAM memory of not more than 1024 MB. Once you create the image and launch it with any of the sample programs, you'll notice a significant increase in the execution of the program. If you want, you may also choose to use GPU acceleration, which will improve the graphics performance by offloading any OpenGL* calls to the host GPU.

INTEL® GRAPHICS PERFORMANCE ANALYZERS (INTEL® GPA)

The Intel GPA Graphics Performance Analyzers are a set of profiling tools that provides metrics and optimization features for graphics intensive applications such as games. One of these tools including the GPA System Analyzer is available for targeting Intel Atom processor-based phones for free and provides a complimentary alternative to the Intel® Vtune™ Amplifier XE so that developers can perform



The Intel Software Manager settings screen